FOR A LOCALLY NOETHERIAN INTEGRAL SCHEME, UFD LOCAL RINGS ARE EQUIVALENT TO NORMALITY TOGETHER WITH SURJECTIVITY OF THE PICARD TO CLASS GROUP MAP

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-local-rings-UFD-c1-bijective is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-divisors/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 6907-6918 and proof span 6920-6953 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof refers to the class group map by the author's equation label and to standard definitions of effective Cartier divisors and their sections; the references are part of the counted span and are handled as navigation pointers.

Editorial note. The second direction uses the same argument twice, once to place an element in the local ring and once to show that every element of the prime is a multiple of it; the author's phrasing is preserved.

Editorial note. 7 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- A unique factorisation domain is normal, and the class group map of a normal integral scheme is injective, both cited from the algebra chapter and the same chapter. Cited at proof lines 6920-6923. Gives normality and injectivity in the first direction.
- A prime divisor on a scheme whose local rings are unique factorisation domains is an effective Cartier divisor, cited from the same chapter. Cited at proof lines 6924-6926. Realizes every prime divisor as a divisor of a section, which gives surjectivity of the class group map.
- An element of the fraction field of a normal Noetherian domain with prescribed orders at height one primes, with order one at a chosen one and order zero elsewhere, lies in the domain, cited from the algebra chapter. Cited at proof lines 6944-6947. Produces a principal generator of each height one prime in the second direction.

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

- A Noetherian normal domain all of whose height one primes are principal is a unique factorisation domain, cited from the algebra chapter. Cited at proof lines 6949-6952. Concludes that the local rings are unique factorisation domains.

Source reference. Original lemma 0AFV: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13173–13173. Same-chapter reference used in the retained author text at source line(s) 6922. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0AFT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13171–13171. Same-chapter reference used in the retained author text at source line(s) 6952. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 031T: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3996–3996. Same-chapter reference used in the retained author text at source line(s) 3417, 6903, 6948, 7032. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original definition 01WX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2498–2498. Same-chapter reference used in the retained author text at source line(s) 3576, 6928, 9257. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original equation 0BE7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14340–14340. Same-chapter reference used in the retained author text at source line(s) 6876, 6910, 6923, 6930, 6935, 7392. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0BE8: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14341–14341. Same-chapter reference used in the retained author text at source line(s) 6924, 7395. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0AGA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 13188–13188. Same-chapter reference used in the retained author text at source line(s) 3153, 3520, 6926. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let X be a locally Noetherian integral scheme. Consider the map (0BE7)* $\mathrm{Pic}(X) \to Cl(X)$*. The following are equivalent*

(1) *the local rings of X are UFDs, and*

(2) *X is normal and* $\mathrm{Pic}(X) \to Cl(X)$ *is surjective.*

In this case $\mathrm{Pic}(X) \to Cl(X)$ *is an isomorphism.*

Proof. If (1) holds, then X is normal by Algebra, Lemma 0AFV. Hence the map (0BE7) is injective by Lemma 0BE8. Moreover, every prime divisor $D \subset X$ is an effective Cartier divisor by Lemma 0AGA. In this case the canonical section 1_D of $\mathcal{O}_X(D)$ (Definition 01WX) vanishes exactly along D and we see that the class of D is the image of $\mathcal{O}_X(D)$ under the map (0BE7). Thus the map is surjective as well.

Assume (2) holds. Pick a prime divisor $D \subset X$. Since (0BE7) is surjective there exists an invertible sheaf $\mathcal{L}$, a regular meromorphic section s, and $f \in R(X)^*$ such that $\mathrm{div}_{\mathcal{L}}(s) + \mathrm{div}(f) = [D]$. In other words, $\mathrm{div}_{\mathcal{L}}(fs) = [D]$. Let $x \in X$ and let $A = \mathcal{O}_{X,x}$. Thus A is a Noetherian local normal domain with fraction field $K = R(X)$. Every height 1 prime of A corresponds to a prime divisor on X and every invertible $\mathcal{O}_X$-module restricts to the trivial invertible module on $\mathrm{Spec}(A)$. It follows that for every height 1 prime $\mathfrak{p} \subset A$ there exists an element $f \in K$ such that $\mathrm{ord}_{A_\mathfrak{p}}(f) = 1$ and $\mathrm{ord}_{A_{\mathfrak{p}'}}(f) = 0$ for every other height one prime $\mathfrak{p}'$. Then $f \in A$ by Algebra, Lemma 031T. Arguing in the same fashion we see that every element $g \in \mathfrak{p}$ is of the form $g = af$ for some $a \in A$. Thus we see that every height one prime ideal of A is principal and A is a UFD by Algebra, Lemma 0AFT. $\square$